

Boss Challenge — Paper 44 (Class 7)

Acids, Bases & Salts | Transportation in Animals & Plants | Rational Numbers | Lines & Angles
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A drop of lemon juice tastes sour and reddens moist blue litmus, marking the class of substances we call:
(A) indicators
(B) salts
(C) acids
(D) bases
2. The liquid that carries digested food, oxygen and wastes around the human body is the:
(A) sweat
(B) bile
(C) saliva
(D) blood
3. If you can express a value as p/q with p and q whole numbers and q not zero, that value is a:
(A) irrational number
(B) whole number
(C) natural number
(D) rational number
4. A pair of angles whose measures together make exactly a right angle of 90° are known as:
(A) supplementary angles
(B) vertically opposite angles
(C) complementary angles
(D) adjacent angles
5. A substance that feels soapy or slippery, tastes bitter, and turns red litmus blue is a:
(A) salt
(B) acid
(C) neutral substance
(D) base
6. Which muscular organ keeps blood moving by pumping it out to every part of the body?
(A) heart
(B) liver
(C) lungs
(D) kidneys
7. On the number line, the rational number $-3/4$ lies:
(A) exactly at -1
(B) between -1 and 0
(C) between -1 and -2
(D) between 0 and 1

8. A pair of angles whose measures together make a straight angle of 180° are described as:
- (A) reflex angles
 - (B) right angles
 - (C) supplementary angles
 - (D) complementary angles
9. Mixing matched amounts of acid with base cancels both natures, and besides a salt the other product made is:
- (A) hydrogen gas
 - (B) more acid
 - (C) water
 - (D) oxygen gas
10. The tubes that carry blood away from the heart to the body are the:
- (A) veins
 - (B) arteries
 - (C) xylem
 - (D) capillaries
11. Which of these is the standard form of the rational number $12/(-18)$?
- (A) $2/3$
 - (B) $-3/2$
 - (C) $-2/3$
 - (D) $-12/18$
12. An angle measures 35° ; the angle needed to complete a right angle with it (its complement) is:
- (A) 65°
 - (B) 35°
 - (C) 145°
 - (D) 55°
13. The chemical that doctors call an antacid works on an upset stomach because it is a mild:
- (A) base that neutralises extra stomach acid
 - (B) salt that has no effect
 - (C) gas that escapes the body
 - (D) acid that adds to stomach acid
14. The tiny thread-like vessels where oxygen and food pass from blood into the cells are the:
- (A) the heart
 - (B) veins
 - (C) arteries
 - (D) capillaries

15. The sum $-\frac{3}{7} + \frac{2}{7}$ equals:

- (A) $\frac{1}{7}$
- (B) $-\frac{1}{7}$
- (C) $-\frac{5}{7}$
- (D) $-\frac{1}{14}$

16. An angle measures 110° ; the angle needed to complete a straight line with it (its supplement) is:

- (A) 20°
- (B) 250°
- (C) 70°
- (D) 90°

17. Turmeric paste is yellow but turns reddish-brown when soap solution is dropped on it. This tells us soap solution is:

- (A) a salt solution
- (B) basic
- (C) acidic
- (D) neutral

18. In plants, the tissue that carries water and minerals up from the roots is the:

- (A) xylem
- (B) arteries
- (C) stomata
- (D) phloem

19. The additive inverse (the number you add to get 0) of $-\frac{5}{9}$ is:

- (A) $\frac{9}{5}$
- (B) $\frac{5}{9}$
- (C) $-\frac{5}{9}$
- (D) 0

20. When two straight lines cross, the angles directly opposite each other are:

- (A) supplementary
- (B) equal
- (C) always 90°
- (D) complementary

21. A gardener finds the soil is too acidic for crops. To raise it toward neutral, the best material to add is:

- (A) lemon juice
- (B) vinegar, which is acidic
- (C) more fertiliser only
- (D) lime, which is basic

22. The plant tissue that carries food made in the leaves to the rest of the plant is the:
- (A) roots
 - (B) phloem
 - (C) xylem
 - (D) veins
23. Multiplying the rational numbers $(-2/3)$ and $(3/4)$ together gives a result of:
- (A) $1/2$
 - (B) $-5/12$
 - (C) $-6/7$
 - (D) $-1/2$
24. Two angles that share a common vertex and a common arm, lying side by side, are called:
- (A) alternate angles
 - (B) adjacent angles
 - (C) complementary angles
 - (D) vertically opposite angles
25. A factory must neutralise its acidic waste with 6 litres of base for every 1 litre of acid. To treat 4 litres of acid it needs:
- (A) 24 litres of base
 - (B) 10 litres of base
 - (C) 6 litres of base
 - (D) 2 litres of base
26. The loss of water as vapour from the surface of leaves is called:
- (A) germination
 - (B) photosynthesis
 - (C) transpiration
 - (D) respiration
27. To compare $2/3$ and $3/5$, the easiest first step is to:
- (A) add the two fractions
 - (B) compare the numerators as they are
 - (C) rewrite both with a common denominator
 - (D) compare the denominators as they are
28. In a linear pair, one angle is 125° . The other angle measures:
- (A) 55°
 - (B) 65°
 - (C) 125°
 - (D) 35°

29. An ant's sting hurts because it injects an acid. The best simple remedy rubbed on the skin is:
- (A) more acid such as vinegar
 - (B) plain table salt
 - (C) a mild base like baking soda
 - (D) lemon juice
30. A resting heart beats about 72 times each minute. In a quarter of a minute it beats about:
- (A) 18 times
 - (B) 24 times
 - (C) 36 times
 - (D) 9 times
31. A diver descends from 0 m to -18 m, then rises 7 m. The diver's new depth (as a rational/integer) is:
- (A) -18 m
 - (B) 11 m
 - (C) -25 m
 - (D) -11 m
32. When a transversal crosses two parallel lines, a pair of corresponding angles are:
- (A) supplementary
 - (B) complementary
 - (C) equal
 - (D) always right angles
33. Phenolphthalein is colourless in acid but turns bright pink in a base. A liquid that turns it pink must be:
- (A) a strong acid
 - (B) basic
 - (C) neutral
 - (D) acidic
34. The rhythmic throb felt in the wrist with each heartbeat is called the:
- (A) pulse
 - (B) reflex
 - (C) hiccup
 - (D) breath
35. Between any two different rational numbers, the count of rational numbers lying between them is:
- (A) infinitely many
 - (B) none
 - (C) exactly ten
 - (D) exactly one

36. A leaning xylem stem makes a 65° angle with the flat ground. The angle it makes with the vertical wall behind it is:

- (A) 65°
- (B) 25°
- (C) 35°
- (D) 115°

37. It takes 5 drops of base to neutralise 1 mL of an acid. A beaker holds 7 mL of the same acid. Drops of base needed:

- (A) 35 drops
- (B) 5 drops
- (C) 70 drops
- (D) 12 drops

38. The waste-removing system that filters the blood and makes urine is the:

- (A) the stomach
- (B) the heart
- (C) excretory system, with the kidneys
- (D) the lungs only

39. The reciprocal (multiplicative inverse) of $-7/4$ is:

- (A) $7/4$
- (B) $4/7$
- (C) $-7/4$
- (D) $-4/7$

40. When a transversal crosses two parallel lines, a pair of alternate interior angles are:

- (A) equal
- (B) complementary
- (C) supplementary
- (D) reflex

41. Litmus is obtained from lichens and is the most common laboratory:

- (A) base
- (B) acid
- (C) indicator
- (D) salt

42. A patient's heart beats 70 times a minute. Over 5 minutes the total number of beats is:

- (A) 75 beats
- (B) 3500 beats
- (C) 350 beats
- (D) 140 beats

43. The value of $(-5/6) - (-1/6)$ is:

- (A) -1
- (B) $2/3$
- (C) $-4/6$
- (D) $-2/3$

44. An angle that is more than 90° but less than 180° is called a/an:

- (A) right angle
- (B) obtuse angle
- (C) reflex angle
- (D) acute angle

45. Which of these is the acid naturally present in lemon and orange juice, giving them their sour taste?

- (A) citric acid
- (B) acetic acid
- (C) hydrochloric acid
- (D) lactic acid

46. Blood is red because it contains a substance that carries oxygen, called:

- (A) plasma
- (B) platelets
- (C) phloem sap
- (D) haemoglobin

47. Which list is written in correct increasing order?

- (A) $1/2, 0, -1/2, -1$
- (B) $0, -1/2, -1, 1/2$
- (C) $-1/2, -1, 0, 1/2$
- (D) $-1, -1/2, 0, 1/2$

48. Two co-interior (same-side interior) angles between parallel lines measure x and 75° . The value of x is:

- (A) 105°
- (B) 285°
- (C) 75°
- (D) 15°

49. Vinegar used in cooking owes its sour taste to a dissolved acid called:

- (A) citric acid
- (B) acetic acid
- (C) tartaric acid
- (D) sulphuric acid

50. Tiny pores on the underside of leaves through which water vapour escapes and gases move are the:
- (A) xylem
 - (B) stomata
 - (C) roots
 - (D) veins
51. Dividing $\frac{4}{9}$ by $\frac{2}{3}$ gives:
- (A) $\frac{3}{2}$
 - (B) $\frac{6}{9}$
 - (C) $\frac{2}{3}$
 - (D) $\frac{8}{27}$
52. Three angles lie on a straight line. Two of them are 50° and 70° . The third angle is:
- (A) 110°
 - (B) 120°
 - (C) 130°
 - (D) 60°
53. A solution leaves both red and blue litmus paper unchanged. The solution is therefore:
- (A) strongly acidic
 - (B) both acidic and basic
 - (C) neutral
 - (D) strongly basic
54. Water rises in a tall tree partly because transpiration at the top creates a:
- (A) pump like an animal heart
 - (B) push from the roots only
 - (C) gas that lifts the water
 - (D) pull that draws water up the xylem
55. A rational number is equal to its own additive inverse only when the number is:
- (A) -1
 - (B) 0
 - (C) 1
 - (D) every rational number
56. Vertically opposite angles formed by two crossing lines are equal. If one such angle is $3x$ and its opposite is 60° , then x is:
- (A) 180
 - (B) 30
 - (C) 20
 - (D) 40

57. When dilute acid is added to a base, the mixture gets warmer. This shows neutralisation:

- (A) absorbs heat from the room
- (B) releases heat energy
- (C) changes no energy at all
- (D) produces light, not heat

58. In animals, the wastes carried by blood to be removed include carbon dioxide and:

- (A) urea
- (B) oxygen
- (C) haemoglobin
- (D) glucose

59. The temperature falls from 5 °C by 8 degrees. Written as a rational/integer, the new temperature is:

- (A) -13 °C
- (B) -3 °C
- (C) 13 °C
- (D) 3 °C

60. Which pair of angles must ALWAYS be equal, no matter the figure?

- (A) adjacent angles
- (B) any two angles on a line
- (C) co-interior angles
- (D) vertically opposite angles

61. A beaker of acid is at 24 °C. Neutralising it with a base raises the temperature by 7 °C. The new temperature is:

- (A) 17 °C
- (B) 28 °C
- (C) 24 °C
- (D) 31 °C

62. If a heart pumps about 70 mL of blood each beat and beats 70 times a minute, blood pumped per minute is about:

- (A) 140 mL
- (B) 700 mL
- (C) 490 mL
- (D) 4900 mL

63. Which number is NOT equal to the others?

- (A) $-\frac{1}{2}$
- (B) $-\frac{2}{4}$
- (C) $-\frac{4}{8}$
- (D) $-\frac{3}{6}$

64. An angle exactly equal to 180° forms a:
- (A) straight angle
 - (B) reflex angle
 - (C) right angle
 - (D) complete angle
65. China rose petals soaked in warm water give an indicator that turns dark pink in acids and green in:
- (A) salt solutions
 - (B) neutral water
 - (C) bases
 - (D) acids
66. Veins carrying blood back to the heart have valves whose job is to:
- (A) stop blood from flowing backward
 - (B) add oxygen to blood
 - (C) speed up the heartbeat
 - (D) make new blood
67. The product of a non-zero rational number and its reciprocal is always:
- (A) 1
 - (B) 0
 - (C) -1
 - (D) the number itself
68. A clock's hands at 3 o'clock form an angle of 90° . This angle is best described as a:
- (A) acute angle
 - (B) right angle
 - (C) reflex angle
 - (D) straight angle
69. Baking soda solution is mildly slippery and turns red litmus blue. Baking soda is therefore a:
- (A) strong acid
 - (B) an indicator
 - (C) neutral salt
 - (D) base
70. Single-celled animals like Amoeba do not need a transport system because materials move by:
- (A) a tiny beating heart
 - (B) its blood vessels
 - (C) simple diffusion across their surface
 - (D) xylem tubes

71. Where does the rational number $7/2$ lie on the number line?

- (A) between 3 and 4
- (B) exactly at 7
- (C) between 7 and 2
- (D) between 2 and 3

72. Two angles of a linear pair are equal. Each angle must measure:

- (A) 90°
- (B) 180°
- (C) 45°
- (D) 60°

73. Acid rain harms buildings and plants. It forms when gases from burning fuels dissolve in:

- (A) the soil to make it neutral
- (B) rainwater to make it acidic
- (C) clouds to make pure water
- (D) rainwater to make it basic

74. The clear, almost colourless liquid part of blood that carries cells and dissolved food is the:

- (A) plasma
- (B) platelets
- (C) urea
- (D) haemoglobin

75. A share price changes by $-2\frac{1}{2}$ points, then by $+1\frac{3}{4}$ points. The total change is:

- (A) $-3/4$ point
- (B) $-1\frac{1}{4}$ points
- (C) $-4\frac{1}{4}$ points
- (D) $3/4$ point

76. A transversal crosses two parallel rails. One corresponding angle is $(2x + 10)^\circ$ and its partner is 70° . Then x equals:

- (A) 60
- (B) 40
- (C) 35
- (D) 30

77. The acid that the human stomach makes to help digest food is:

- (A) carbonic acid
- (B) citric acid
- (C) hydrochloric acid
- (D) acetic acid

78. A plant absorbs water mainly through fine outgrowths of its roots called:
- (A) petals
 - (B) stomata
 - (C) root hairs
 - (D) phloem
79. Which statement about rational numbers is true?
- (A) Every rational number is an integer
 - (B) Rational numbers cannot be negative
 - (C) No integer is a rational number
 - (D) Every integer is a rational number
80. An acute angle is one that measures:
- (A) less than 90°
 - (B) more than 90°
 - (C) exactly 90°
 - (D) exactly 180°
81. The product formed when an acid neutralises a base, apart from water, is always a:
- (A) another acid
 - (B) metal
 - (C) gas
 - (D) salt
82. A nurse counts 18 heartbeats in 15 seconds. The heart rate per full minute is:
- (A) 72 beats per minute
 - (B) 90 beats per minute
 - (C) 33 beats per minute
 - (D) 18 beats per minute
83. The value of $0 \div (-5/8)$ is:
- (A) undefined
 - (B) $-5/8$
 - (C) $-8/5$
 - (D) 0
84. Around a single point, all the angles together add up to:
- (A) 270°
 - (B) 90°
 - (C) 180°
 - (D) 360°
85. Soap solution turns red litmus blue. If you then add lemon juice drop by drop, the blue colour will:
- (A) turn colourless
 - (B) fade back toward red as the acid neutralises the base
 - (C) turn green
 - (D) stay blue forever

86. Blood that has given up its oxygen to the body and is now carrying carbon dioxide is described as:
- (A) deoxygenated
 - (B) digested
 - (C) filtered
 - (D) oxygenated
87. The rational number that is neither positive nor negative is:
- (A) $\frac{1}{2}$
 - (B) 0
 - (C) -1
 - (D) 1
88. Two acute angles measuring 40° and 35° are placed together at a point with a 285° angle. Do they complete a full turn?
- (A) yes, because they add to 90°
 - (B) no, they add to 300°
 - (C) yes, because $40 + 35 + 285 = 360$
 - (D) no, they add to 180°
89. Among water, milk of magnesia, lemon juice and vinegar, the only clearly BASIC one is:
- (A) milk of magnesia
 - (B) lemon juice
 - (C) vinegar
 - (D) water
90. Why must large animals have a transport system while a tiny Amoeba does not?
- (A) diffusion is faster in large animals
 - (B) their cells are too far from the surface for diffusion alone
 - (C) small animals need more oxygen
 - (D) they have no cells that need oxygen
91. A recipe needs $\frac{3}{4}$ cup of milk per serving. For $\frac{2}{3}$ of a serving you need:
- (A) $\frac{1}{2}$ cup
 - (B) $\frac{3}{4}$ cup
 - (C) $\frac{5}{12}$ cup
 - (D) $\frac{9}{8}$ cup
92. Which statement is TRUE about supplementary angles?
- (A) Supplementary angles must be equal
 - (B) Both supplementary angles are always acute
 - (C) Supplementary angles always add to 90°
 - (D) Two supplementary angles cannot both be acute

93. Two acidic drinks measure 'sourness' 3 and 5 on a sour-scale, and a base measures -4. Adding the base to the '5' drink moves its reading to:
- (A) 9
 - (B) 0
 - (C) 1
 - (D) -1
94. Over a 60-second minute the heart rests briefly between beats, beating 75 times. The average time for one full beat is:
- (A) 8 seconds
 - (B) 0.08 seconds
 - (C) 0.8 seconds
 - (D) 1.25 seconds
95. Which property is shown by $(-2/5) + (3/7) = (3/7) + (-2/5)$?
- (A) multiplication is commutative
 - (B) 0 is the additive identity
 - (C) addition of rational numbers is commutative
 - (D) addition is associative
96. A beam of light strikes a mirror so the angle between the beam and the mirror is 30° . The angle between the beam and the line perpendicular to the mirror (the normal) is:
- (A) 120°
 - (B) 30°
 - (C) 150°
 - (D) 60°
97. Which statement correctly pairs a property with the right type of substance?
- (A) Acids feel slippery and turn red litmus blue
 - (B) Bases feel slippery and turn red litmus blue
 - (C) Bases are sour and turn blue litmus red
 - (D) Acids are bitter and turn red litmus blue
98. Which correctly matches the transport pathway to what it carries?
- (A) Xylem carries food; phloem carries water
 - (B) Xylem carries water up; phloem carries food
 - (C) Phloem carries water up from the roots
 - (D) Both xylem and phloem carry only water
99. A submarine at -40 m below sea level rises in 5 equal stages to the surface (0 m). Each stage changes its depth by:
- (A) -8 m
 - (B) +40 m
 - (C) +8 m
 - (D) +5 m

100. Two parallel branches of a plant are crossed by a slanting twig. One angle made is 65° . The co-interior angle on the same side measures:

- (A) 295°
- (B) 115°
- (C) 65°
- (D) 25°